

Simple Angle Chasing

Notations

H is the orthocenter. I is the Incenter. O is the circumcenter. They are all centers of triangle ABC.

Orange and dotted objects are what we wish to show/ find.

Circular dot denotes center of a circle.

Elementary tools for angle chasing

1. Triangle Sum: the sum of the angles in a triangle is 180°
2. Parallel lines preserve angles
3. Circles preserve angles
4. Circles also preserve opposite angles, with 180°
5. Angle at center doubles

Triangle Sum

- Triangle Sum is fundamental. There is no 'best way' nor tricks to use it, this would be saying how to use multiplication correctly.
- The exercises aim to help the readers to recall triangle sum is , in fact, a property that can be used.

Example 1 Represent $\angle BIC$ in terms of $\angle A$

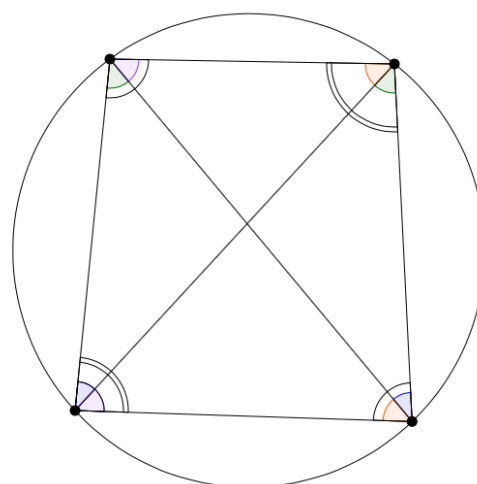
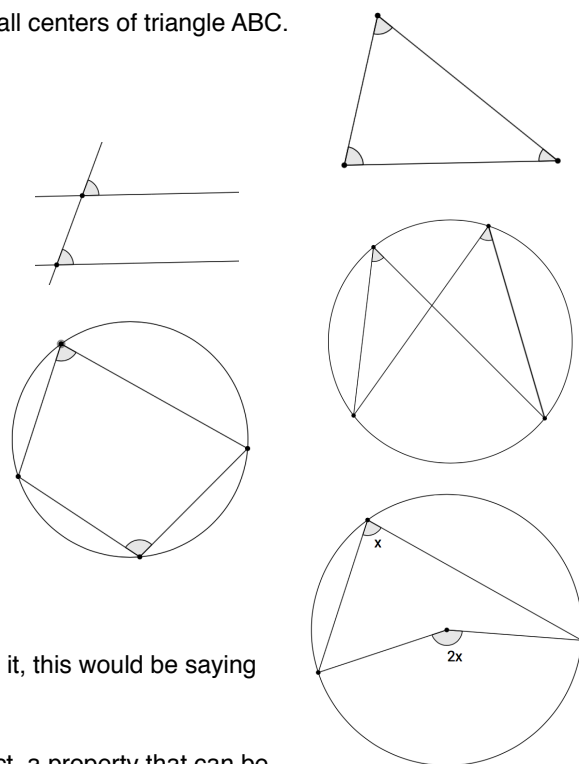
Problem 2 Represent $\angle BHC$ in terms of $\angle A$

Problem 3 Represent $\angle BOC$ in terms of $\angle A$, hence or otherwise

Problem 4 Show that $\angle BAH = \angle CAO$

Circles

- A concyclic relationship implies 6 pairs of angle relationships. Any one of the 6 can imply the concyclicity. Therefore, proving concyclic points gives 5 more equal (directed) angles.
- Concyclic points are significantly more powerful than fundamental techniques, such as triangle sum. The power comes from the unintuitive constructions of a center with 4 equal edges.
- Consequently, knowing which one of the 6 pairs to prove is essential.
- Advice: Try all 6



Example 5 Two circles intersect at two points A,B. A line intersect the two circles at S,T and P,Q respectively. Show $\angle SAP = \angle QBT$

Example 6 X,Y,Z are points on BC, CA, AB respectively. Circumcircle of AZY and BXZ intersect at P. Show that PYCX is cyclic.

Problem 7 ABCD is a cyclic quadrilateral, E, F are points on line AB, CD such that $EF \parallel BC$, show that AEFD is cyclic.

Right angles

- Two right angles sum up to 180 degrees, giving us four concyclic points immediately.
- If there are multiple right angles, it would be a good idea to spot the corresponding concyclic points. Sometimes, it may be necessary to construct extra right angles to make use of these conditions.

Example 8 Show that $\triangle AH_BH_C$ is similar to $\triangle ABC$, where H_B, H_C denote the feet of altitudes.

Example 9 $\angle MPA = \angle MQA = 90^\circ$, the feet of perpendicular from A to PQ is K. Show that $\angle PAK = \angle MAQ$

Problem 10 A circle is divided into equal arcs by n diameters. Prove that the feet of the perpendiculars dropped from an arbitrary point M inside the circle to these diameters are vertices of a regular n-gon.

Problem 11 Show that H is the incenter of $H_AH_BH_C$, where H_A, H_B, H_C are feet of altitude.

Problem 12 Show that the feet of perpendicular from H_A to AC, AB, B-Altitude, C-Altitude, are all collinear

Angle chasing concepts

Useful Angles

- A useful angle is those that show promises to be angle chased.
- If a certain angle requires connection across points that have multiple constructions in-between, then it is likely not useful, unless with support of other concyclic conditions.
- Without loss of generality, if there are two symmetrically constructed points, we may consider only the two angle relationships with similar angles.

Example 13 From a quadrilateral construct four external angle bisectors, adjacent angle bisector intersects at four points. Show that these four points are concyclic.

Example 14 Show that $BICE_A$ is cyclic, with the center lying on circumcircle of ABC. E_A is the Ex-center opposite to vertex A.

Non-useful angles

- Some angles are impossible to be chased using only similar triangles and concyclic points.
- Some concyclic relationships are impossible to be proven by angle chasing.

Example 15 Let ABC be an acute triangle with D, E, F the feet of the altitudes lying on BC, CA, AB respectively. One of the intersection points of the line EF and the circumcircle is P. The lines BP and DF meet at point Q. Prove that PQAF is cyclic. [Level 1, 10G1]

Problem 16 Two circles intersect at AB, lines through A intersect the two circles at P,Q. Line through B intersect the two circles at S,T, then $PS \parallel QT$

Problem 17 Prove that reflection of H in BC lies on circumcircle of ABC

Problem 18 Let B be a point on a circle S_1 , and let A be a point distinct from B on the tangent at B to S_1 . Let C be a point not on S_1 such that the line segment AC meets S_1 at two distinct points. Let S_2 be the circle touching AC at C and touching S_1 at a point D on the opposite side of AC from B . Prove that the circumcentre of triangle BCD lies on the circumcircle of triangle ABC . [Level 1, 02G1]

Feel before Chase

- Before actually evaluating the angles, one should use his intuitions to evaluate whether or not an attempt to chase an angle will be successful.

- Angles extend along a line. Angles about lines with three collinear points are more likely to be useful.

Example 19 AD is the angle bisector of angle ABC , where D lies on BC . The tangent at A of circumcircle of ABC meets BC at P . Show that $PA = PD$

Example 20 AB is a chord on circle Γ with center O . C is a point on circumcircle of ABO . AC meets Γ again at D . Show that DCB is isosceles.

Problem 21 Two circles meet at A, B . O_1A extended meet the circle 2 at C . O_2A extended meet circle 1 at D . Show that C, D, O_1, O_2, B are concyclic,

Problem 22 Two circles are externally tangent at A . B, C , are points on the two circles respectively. Tangents of B, C at their respective circles meet at D . O is the circumcenter of ABC . Show that $ODBC$ is cyclic.

Problem 23 Let Γ be the circumcircle of triangle BOC . Γ intersects AB again at P , AC again at Q . ON is a diameter of Γ . Prove that $APNQ$ is a parallelogram.

Problem 24 AE is a diameter of circumcircle of ABC . AC intersect tangent at B at K . Line through K perpendicular to AE intersects circumcircle of BCK again at D . Show that CE bisects $\angle BCD$.

Angle-Splitting

- In order to evaluate a not-so-useful angle, express it in terms of sum or differences of two or more rather-useful angles.

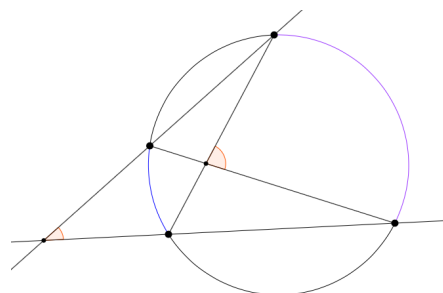
Example 25 Two circles intersect at two points A, B , a line through B intersect two circles at C, D . Tangents at C, D to their respective circles meet at E . Show that $ACDE$ concyclic.

Problem 26 Four lines intersect at 6 points $A_{12}, A_{13}, A_{14}, A_{23}, A_{24}, A_{34}$. Circumcircle of $A_{12}A_{13}A_{23}$ and $A_{12}A_{14}A_{24}$ intersect at P . Show that P also lies on circumcircle of $A_{13}A_{14}A_{34}$ and $A_{23}A_{24}A_{34}$.

Problem 27 $ABCD$ cyclic. Let E be the intersection of lines parallel to AC and BD passing through B and A , respectively. The lines EC and ED intersect the circumcircle of AEB again at F and G , respectively. Prove that points C, D, F and G lie on a circle.

Arc Chasing

- Instead of angles as drawn, regard arcs on a circle as the basic unit of angles
- If most angle chasing actions occur in a circle, then arc chasing can simplify and give great insight.
- Arc chasing lemma can shorten angle chasing:
Interior intersection = Sum of two arcs
Exterior intersection = Difference of two arcs



Example 4' Show that $\angle BAH = \angle CAO$

Example 28 In a circle, M is the mid-point of arc BC , two rays from M intersect chord BC and the circle again at four distinct points. Show that these four points are concyclic.

Example 29 $ABCD$ has an incircle. The four tangent points form a quadrilateral. Show that the four midpoints of this quadrilateral is a rectangle iff $ABCD$ is cyclic.

Problem 30 $ABCD$ is a square. M is a point on minor arc DC . AM intersect BD, CD at P, R respectively. BM intersect AC and DC at Q, S respectively. Show that $PS \perp QR$

Problem 31 Let P be a point interior to triangle ABC (with $CA \neq CB$). The lines AP, BP and CP meet again its circumcircle Γ at K, L , respectively M . The tangent line at C to Γ meets the line AB at S . Show that from $SC = SP$ follows $MK = ML$. [Level 2, 10G2]

(Hint: You need something besides angle chasing)

Problem solving with angle chasing

General geometry problem-solving techniques

- Deduce, in one way or the other, some extra properties of the given points. They can link the given conditions to the required result.

Technique 1: Using the given conditions and randomly collide them in order to produce some meaningful result

Example 32 ABC has $\angle A = 90^\circ$ and $AB = AC$. D is a point on BC . P, Q are projections from D to AB, AC . Show that the reflection of D in PQ lies on circumcircle of ABC .

Problem 33 $ABCD$ is a parallelogram. P is a point on BD such that $\angle PCB = \angle ACD$. Circumcircle of $\triangle ABD$ intersect AC at E . Prove that $\angle AED = \angle PEB$

Technique 2: Observe from the diagram / Make a random educated guess

Example 34 Tangents at B, C to circumcircle of ABC meet at P . The circle through B and tangents AC at A , intersect with the circle through C and tangents AB at A at D . Show that A, D, P collinear.

Problem 35 B_1, C_1 are the projections of B, C on AC, AB respectively. B_2, B_3 are projections of B_1 on AC, BC respectively. C_2, C_3 are projections of C_1 on AB, BC respectively. Show that $B_2B_3 = C_2C_3$.

Technique 3: Using the final condition, deduce what has to be true for the final condition to hold.

At the same time, this will simplify the problem statement.

Example 36 Two circles intersect at A,B. A line intersect two circles at A_1, A_2 and B_1, B_2 respectively. Show that $\angle A_1 A B_1 = \angle A_2 A B_2$

Problem 37 Let ABC be a triangle, and let D be a point on side BC. A line through D intersects side AB at X and ray AC at Y. The circumcircle of triangle BXD intersects the circumcircle ω of triangle ABC again at point Z. ZD and ZY intersect ω again at V and W respectively. Prove that $AB=VW$.

Extra technique: Construct points or lines.

- Not applicable in angle chasing

Non-Angle Chasing Lemmas

Theorem 38 (Converse of semi-circle) $\angle A = 90^\circ$, M is the mid point of BC. Show that $MA = MB = MC$

Example 39 D and E are midpoints of AB and AC. The internal angle bisector of $\angle ABC$ intersect DE at P. Prove that $AP \perp BP$.

Theorem 40 (Mid-point of arc) The angle bisector of $\angle A$ meets perpendicular bisector of BC at D. Show that ABCD is cyclic

Theorem 41 (Isosceles Trapezium) Reflect A along perpendicular bisector of BC to get A' , show that $AA'BC$ is cyclic.

Try to apply these three techniques. You may have to use more than one, or none at all. If this is predictable, it would not be IMO.

Problem 42 Points E and F are chosen on the side AB of a convex quadrilateral ABCD such that $AE < AF$. Given that $\angle ADE = \angle FCB$ and $\angle EDF = \angle ECF$, prove that $\angle FDB = \angle ACE$.

Problem 43 l is a line passing through A. P is projection from A to BC. E, F are projection from B, C on l . M, N are the midpoints of AB, AC. EM, FN intersect at P. Show that DEFP is cyclic.

Problem 44 Let ABC be an acute angled triangle with $\angle B = 60^\circ$ and $AB > AC$. Prove that $2\angle AHI = 3\angle ABC$

Problem 45 P is a point such that $PB \perp CB, PA = PB$. Q is a point such that $QC \perp CB, QC = QA$. O_1 is the circumcircle of OPQ. Show that $\angle O_1AO = 90^\circ$

Problem 46 A, B_1, C_1 and A, C_2, B_2 are collinear in this order, with $B_1C_1 = C_2B_2$. Circumcircle of AB_1B_2 and AC_2B_2 intersect at P. Show that AP bisects $\angle BAC$.

Problem 47 Two circles intersect at A,B. A line through B intersect circle 1 at K, circle 2 at M. Q is a point on circle 1 such that tangent at Q is parallel to AM. AQ intersect circle 2 again at R. Prove that the tangents at Q, R, and KM concurrent.

Problem 48 Angle bisector of $\angle A$ meets BC at D, and circumcircle of ABC at P. Line perpendicular to AB through D intersect AB at E, circumcircle at Q. PQ intersect CB at F. CQ intersect AF at G. Show that $EG \parallel BC$.

Problem 49 $\triangle ABC$ has $\angle A = 60^\circ$. E, F are points on extensions of AB, AC such that $BE = CF = BC$. Circumcircle of ACE intersect FE again at K. Show that AK bisects $\angle A$.

Problem 50 Two circles are such that O_1 lies on circle 2. Two circles intersect at A,B. O_1O_2 intersect circle 1 at C, BC intersect circle 2 again at D. DA intersect circle 1 again at E. F is the projection from O_1 to AE. Show that O_1A, O_1D trisect

$\angle CO_1F$.

Problem 51 AI meets BC at D. The perpendicular bisector of AD intersect angle bisectors of $\angle B, \angle C$ at P, Q respectively. Show that P, Q, A, I are concyclic.

Problem 52 P, Q are projections from H_A to AB, BC . Show that PQ bisects H_AH_B

Advanced Applications of Angle Chasing

Transform conditions into an angle chasing equivalent

Example 53 Points A,B,M and N on a circle is given. Chords MA_1, MB_1 are perpendicular to lines NB, NA respectively. Prove that $AA_1 \parallel BB_1$.

Problem 54 ABCD is cyclic. AB and CD meet at E. AD and CB meet at F. Internal angle bisector of $\angle F$ intersect DC at R and AB at P. internal angle bisector of $\angle E$ intersect AD and CB at S and Q. Show that RQPS is a rhombus.

Example 55 ABCD cyclic. Diagonals intersect at E. O_1, O_2, O are circumcenters of AED, BEC, ABCD respectively. Show that EO_1OO_2 is a parallelogram.

Problem 56 Triangle ABC has circumcircle Ω and circumcenter O . A circle Γ with center A intersects the segment BC at points D and E, such that B, D, E, and C are all different and lie on line BC in this order. Let F and G be the points of intersection of Γ and Ω , such that A, F, B, C, and G lie on Ω in this order. Let K be the second point of intersection of the circumcircle of triangle BDF and the segment AB. Let L be the second point of intersection of the circumcircle of triangle CGE and the segment CA.

Suppose that the lines FK and GL are different and intersect at the point X. Prove that X lies on the line AO. [Level 1, 15G2 (P4)]

Problem 57 The tangents at B,C, to the circumcircle ABC intersects tangent at A at K,L. Line through K parallel to AB intersect line through L parallel to AC at P. Show that $BP = CP$

Exercises

Problem 58 Prove that A' , the reflection of H in mid point of BC lies on circumcircle of ABC, and furthermore $A'A$ is a diameter.

Problem 59 AE is a diameter of circumcircle of ABC. AC intersect tangent at B at K. Line through K perpendicular to AE intersects circumcircle of BCK again at D. Show that CE bisects $\angle BCD$

Problem 60 Let ABC be a triangle with orthocenter H and let M be the midpoint of BC. ω_B is the circumcircle of B, H, M. ω_C is the circumcircle of C, H, M. Lines AB and AC meet ω_B and ω_C again at P and Q, respectively. Rays PH and QH meet ω_C, ω_B again at R,S respectively. Prove that M,R,S are collinear

Problem 61 The sides of a Pentagon is extended to form a pentagram. Adjacent circumcircles of the 5 newly formed triangles intersect at 5 points. Prove that these 5 points are concyclic.

Problem 62 $\triangle ABC$ has $\angle A = 60^\circ$. E,F are points on extensions of AB,AC such that $BE = CF = BC$. Circumcircle of ACE intersect FE again at K. Show that AK bisects $\angle A$.

Problem 63 Let ABC be an acute-angled triangle with $AB \neq AC$. The circle with diameter BC intersects the sides AB and AC at M and N respectively. Denote by O the midpoint of the side BC. The bisectors of the angles $\angle BAC$ and $\angle MON$

intersect at R . Prove that the circumcircles of the triangles BMR and CNR have a common point lying on the side BC .
[Level 1, 04G1]

Problem 64 Let ABC be an acute triangle with D, E, F the feet of the altitudes lying on BC, CA, AB respectively. One of the intersection points of the line EF and the circumcircle is P . The lines BP and DF meet at point Q . Prove that $AP = AQ$. [Level 1, 10G1]

Problem 65 Given triangle ABC the point J is the centre of the excircle opposite the vertex A . This excircle is tangent to the side BC at M , and to the lines AB and AC at K and L , respectively. The lines LM and BJ meet at F , and the lines KM and CJ meet at G . Let S be the point of intersection of the lines AF and BC , and let T be the point of intersection of the lines AG and BC . Prove that M is the midpoint of ST . [Level 1, 12G1]

Problem 66 Denote by M midpoint of side BC in an isosceles triangle $\triangle ABC$ with $AC = AB$. Take a point X on a smaller arc $\widehat{MA}$ of circumcircle of triangle $\triangle ABM$. Denote by T point inside of angle BMA such that $\angle TMX = 90^\circ$ and $TX = BX$. Prove that $\angle MTB - \angle CTM$ does not depend on choice of X . [Level 1, 07G2]

Problem 67 Let ABC be an acute triangle with D, E, F the feet of the altitudes lying on BC, CA, AB respectively. One of the intersection points of the line EF and the circumcircle is P . The lines BP and DF meet at point Q . Prove that $AP = AQ$. [Level 1, 10G1]

Problem 68 Triangle BCF has a right angle at B . Let A be the point on line CF such that $FA = FB$ and F lies between A and C . Point D is chosen so that $DA = DC$ and AC is the bisector of $\angle DAB$. Point E is chosen so that $EA = ED$ and AD is the bisector of $\angle EAC$. Let M be the midpoint of CF . Let X be the point such that $AMXE$ is a parallelogram. Prove that BD, FX and ME are concurrent. [Level 1, 16G1(P1)]

Problem 69 Let $A_1A_2A_3\dots A_n$ be a regular n -gon. Let B_1 and B_n be the midpoints of its sides A_1A_2 and $A_{n-1}A_n$. Also, for every $i \in \{2, 3, 4, \dots, n-1\}$. Let S be the point of intersection of the lines A_1A_{i+1} and A_nA_i , and let B_i be the point of intersection of the angle bisector of the angle $\angle A_iSA_{i+1}$ with the segment A_iA_{i+1} . Prove that $\sum_{i=1}^{n-1} \angle A_1B_iA_n = 180^\circ$.

[Level -, 04G-]

Problem 70 Points A_1, B_1, C_1 are chosen on the sides BC, CA, AB of a triangle ABC respectively. The circumcircles of triangles $AB_1C_1, BC_1A_1, CA_1B_1$ intersect the circumcircle of triangle ABC again at points A_2, B_2, C_2 respectively ($A_2 \neq A, B_2 \neq B, C_2 \neq C$). Points A_3, B_3, C_3 are symmetric to A_1, B_1, C_1 with respect to the midpoints of the sides BC, CA, AB respectively. Prove that the triangles $A_2B_2C_2$ and $A_3B_3C_3$ are similar. [Level -, 06G-]

Problem 71 Let $ABCD$ be a circumscribed quadrilateral. Let g be a line through A which meets the segment BC in M and the line CD in N . Denote by I_1, I_2 and I_3 the incenters of $\triangle ABM, \triangle MNC$ and $\triangle NDA$, respectively. Prove that the orthocenter of $\triangle I_1I_2I_3$ lies on g .
[Level -, 09G-]

Problem 72 Let ABC be an acute triangle with circumcircle Γ . Let ℓ be a tangent line to Γ , and let ℓ_a, ℓ_b and ℓ_c be the lines obtained by reflecting ℓ in the lines BC, CA and AB , respectively. Show that the circumcircle of the triangle determined by the lines ℓ_a, ℓ_b and ℓ_c is tangent to the circle Γ .
[Level -, 11G-(P-)]